

Chapter 97 — Surgical Management of Ureteral Calculi

- **Lifetime risk of a ureteral stone is 12% in males and 6% in females**, so every urologist manages the problem.
- **Obstruction distends the renal capsule, which causes the pain**; the dangerous consequences are **renal damage and infection or sepsis**.
- Depending on size, location and circumstances, **spontaneous passage may be attempted first**.
- **Diagnosis can be difficult** because the pain mimics conditions from **appendicitis to ruptured aortic aneurysm**; imaging is the most definitive step.
- **Absence of hydronephrosis alone does not exclude a ureteral stone**. Ultrasound is now argued by its proponents to be almost as sensitive as CT.

IMAGING FOR SUSPECTED URETERAL STONE — COMPARISON

Modality	Strength	Limitation
Noncontrast computed tomography (NCCT)	Gold standard; highest sensitivity; has replaced plain radiography and intravenous pyelography; low-dose protocols are as effective as full dose	Ionising radiation
Ultrasonography (US)	No ionising radiation	Misses stones between the ureteropelvic junction (UPJ) and ureterovesical junction (UVJ), i.e. the mid-ureter
US + kidney-ureter-bladder radiograph (KUB)	Higher combined sensitivity; performs better once a prior NCCT has localised the stone	Radiolucent stones invisible on KUB

MEDICAL EXPULSIVE THERAPY

- **Medical expulsive therapy (MET) with α -blockers**, which relax ureteral smooth muscle, has been the mainstay of adjunctive treatment.
- **MET remains controversial**: the large prospective **SUSPEND trial (Pickard et al., 2015)** showed **no difference in the need for intervention at 4 weeks and equivalent passage rates** versus control. Publications nonetheless continue on **α -blockers, calcium channel blockers and phosphodiesterase-5 inhibitors**.
- **Spontaneous passage becomes less likely with time**, so **intervention should generally be considered after 4–6 weeks**.
- **Interventions from least to most invasive**: shock wave lithotripsy (SWL) → ureteroscopy (URS) → percutaneous nephrolithotomy (PCNL) with antegrade URS.

URGENCY OF INTERVENTION — COMPARISON

Tier	Trigger
Elective	Failed trial of spontaneous passage
Urgent	Unrelenting pain; renal dysfunction secondary to obstruction
Extremely urgent	Obstructive pyelonephritis or sepsis

Preparation of the Patient Before Surgery

- **History and examination identify absolute contraindications to PCNL or SWL**, such as **uncorrected coagulopathy**, and any **active, untreated urinary tract infection (UTI)**.

- **An obstructed, infected kidney is drained only** — percutaneous nephrostomy or ureteral stent — **without manipulating the stone.**
- **Laboratory tests:** full blood count, electrolytes, renal function, and a **urine culture so that perioperative antibiotics can be tailored to the organism.**
- **Every patient needs preoperative imaging.** Before SWL a **KUB confirms the stone is radio-opaque** (unless US-guided SWL is available); a preoperative KUB also allows **follow-up by KUB and US instead of repeat NCCT**, reducing radiation.

ANTITHROMBOTIC MANAGEMENT BY MODALITY — COMPARISON

Modality	Antiplatelet or anticoagulant	Bridging
PCNL	Stop if medically feasible	Short-acting heparin if thrombotic risk is high
SWL	Stop if medically feasible	Not recommended (too few data); if bridging is needed, choose another modality
URS with laser	Need not be stopped	Not required; expect more gross haematuria and worse stent symptoms

Concept — Antithrombotic policy follows the energy: SWL and PCNL injure parenchyma where bleeding cannot be seen or controlled, whereas laser URS works under direct vision within the lumen, so only URS tolerates continued anticoagulation.

TREATMENT DECISION

FACTORS IN CHOOSING TREATMENT FOR URETERAL CALCULI

Domain	Factors
Surgeon	Experience, training, skills, resources
Patient	Symptoms, expectations, comorbidities, anatomy; body habitus and skin-to-stone distance (SSD)
Stone	Location, size, density, radio-opacity, composition, presence and duration of obstruction, infection
Technical	Equipment, cost, anaesthetic support, health-system resources

- Surgeons need **several techniques at their disposal** to achieve the best outcome.
- **SWL copes poorly with larger stones** because the patient must pass many fragments or undergo repeated sessions; the **historical 2 cm upper limit** has given way to evidence of **poorer stone-free rates (SFRs) above 10–15 mm.**
- **A plain-language shared decision aid for URS versus SWL** both informs the patient about the procedures and informs the urologist of that patient's goals.
- **Impaction is gauged on CT or US by the thickness of the ureteral wall around the stone.** Severely impacted stones resist SWL and are better treated endoscopically, retrograde or antegrade.
- **Both SWL and URS can treat a stone anywhere in the ureter.** In the **upper ureter (≤ 2 cm)** a randomised trial found **similar SFR at 3 months, more retreatment with SWL and similar complications**; in the **distal ureter**, SFRs are similar, SWL more often needs a second session, and **complications are lower with SWL.**
- **Large, hard proximal stones favour PCNL with antegrade URS**; in a **comorbid patient unfit for PCNL**, repeated retrograde URS may be safer.

SIZE CEILINGS BY MODALITY — COMPARISON

Modality	Size guidance
SWL	Historical limit 2 cm; results decline above 10–15 mm
URS	No definitive limit

Modality	Size guidance
PCNL ($\pm$ antegrade URS)	Consider above 15–20 mm, especially large, hard proximal stones

Concept — Size sets the ceiling for each modality, and impaction and SSD erode SWL further; within the ceilings the trade is constant — SWL gives fewer complications but more sessions; URS gives one session at slightly higher morbidity.

URETEROSCOPE CHOICE

- Three types exist — **rigid, semirigid and flexible** — differing in size, optics and working channels. Availability often dictates the choice, but anatomy, stone location and expertise matter. Scopes were designed as **reusable**; **single-use flexible scopes** are recent.

Semirigid Ureteroscopes

- **Rigid scopes use the Hopkins rod-lens system**: contoured glass rods with intervening air spaces acting as lenses. The **outer diameter increases toward the base**, so the scope itself dilates the orifice and ureter.
- **Force applied off the scope's axis distorts the rod-lens image**, first seen as a **semicircular blacked-out area**.
- The **semirigid scope (1989)** replaced the rod lenses with a **flexible fused-quartz fibre bundle** inside a rigid casing, allowing flexion of **up to 2 inches off axis**; it rapidly superseded rigid scopes.
- **Semirigid scopes are robust and cheap**, but are damaged by **torque** — **typically at the pelvic brim, where the shaft can bend or break** — and by **laser fired too close to the tip**.

Flexible Ureteroscopes

- **Flexible scopes suit upper ureteral and renal stones**: if a proximal stone migrates into the kidney, the scope can follow and treat it.
- **Deflection up to 270° reaches even the lower pole**, and the **standard 3.6-French (Fr) working channel** admits small baskets or laser fibres while preserving irrigation.
- Fibreoptic and digital, reusable and single-use versions exist, and **no single type is best for all cases**.

Fibreoptic Flexible Ureteroscopes

- The **fibre bundles that carry light and image are delicate**; deflection, torque and bending break fibres early in a scope's life, producing **black spots and falling brightness** until repair is needed.
- **Channel leaks show as lens fogging or water staining** and demand repair, because **catastrophic failure usually follows during reprocessing**.
- Bundle density varies by manufacturer; a **honeycomb pattern limits image fidelity, worse with coarser bundles**.

Digital Flexible Ureteroscopes

- Introduced in the **mid-2000s** with **charge-coupled device (CCD) or complementary metal-oxide semiconductor (CMOS)** sensors that convert photons to electrons at the tip ("**chip-on-the-tip**"). Because the signal travels digitally, **the image does not degrade over time**.
- **CMOS runs cooler, uses less energy, processes faster and is cheaper than CCD**, so CCD has been phased out.
- **No camera head and fewer cords improve ergonomics**.
- In comparison studies digital scopes gave **better clarity, magnification and manoeuvrability without moiré**, reached the whole collecting system more often, and **shortened operative time with similar SFRs**. The better view prompted speculation about seeing the **earliest stages of calcium oxalate stone formation**.

- **Fibreoptic scopes survive because the chip makes the digital tip larger** — about 8.5 Fr, flatter and less tapered — so entering the orifice and a narrow ureter is harder. Future miniaturisation may close this gap; meanwhile **all major manufacturers still make both fibreoptic and digital scopes**.

Durability and Failure Rates of Flexible Ureteroscopes

- **Fibreoptic scopes have always needed frequent repair**, and modern designs have not improved on earlier generations.
- **Scope-sparing technique extends the repair interval**: relocating lower-pole stones to an accessible calyx, using **small-core (200 µm) fibres**, working through a **ureteral access sheath (UAS)** and avoiding deflection inside it.
- **Digital scopes were hoped to be more durable**, and early single-centre series suggested so, but **later comparison showed failure rates similar to fibreoptic scopes**.
- **Keep a fibreoptic scope available**: when a digital scope cannot pass a **narrow UPJ or infundibulum**, the smaller fibreoptic scope has reached the target, sparing a second procedure.
- **Overall, one repair falls due roughly every 15 procedures**, at high cost per repair — a burden on health systems.
- **Most centres therefore run a hybrid fleet**: reusable scopes routinely, **single-use scopes for difficult cases likely to damage an instrument**.

Single-Use Digital Flexible Ureteroscopes

- **Single-use digital flexible ureteroscopes (SDFUs) answer the repair problem**: a new scope with **maximum deflection every case, no reprocessing cost or staff**, and a **fixed, budgetable cost** instead of unpredictable repairs.
- **Bench, cadaveric and clinical performance matches reusable scopes**, so **cost-effectiveness becomes the deciding question**. The true cost of a reusable scope includes purchase, repair, cleaning, sterilisation, quality checks and lost opportunity when it fails; micro-costing has found **similar cost per case**.
- **Economics shift with infrastructure** — a new centre without towers or processors faces higher capital cost — so **each centre must do its own analysis**.
- Later SDFUs are **all CMOS-based** with subtle differences and **few direct comparisons**; manufacturers are prioritising single-use development.

LITHOVUE (BOSTON SCIENTIFIC, 2016) — FIRST WIDELY ADOPTED SDFU

Feature	Specification
Sensor; field of view	CMOS; 2–50 mm
Diameter	7.7 Fr tip, 9.5 Fr shaft
Light source	Integrated light-emitting diode (LED) ; no external light source
Working channel; deflection	Single 3.6 Fr channel at 3 o'clock; ≥270° in both directions
Connection; sterility	Single cable to a dedicated processor; supplied sterile, not designed or approved for reprocessing

URETEROSCOPE TYPES — COMPARISON

	Semirigid	Fibreoptic flexible	Digital flexible (reusable)	Single-use digital
Image over time	Stable unless shaft bent	Degrades as fibres break	No degradation	New each case
Calibre	Tapered; dilates orifice	Smallest, tapered tip	Larger, flatter tip	7.7 Fr tip
Platform	Standard	Agnostic	Proprietary processor only	Dedicated processor

	Semirigid	Fibreoptic flexible	Digital flexible (reusable)	Single-use digital
Cost logic	Robust, cheap	Repair about every 15 cases; reusable cheaper above about 99 cases per year		Cheaper below about 99 cases per year

The table tests **calibre against image** and **volume against cost**. The trap is assuming the digital scope is more durable or universally better: it is neither, and its proprietary processor matters when equipping a theatre.

Intrarenal Pressure Monitoring

- **Raised intrarenal pressure (IRP) drives fluid absorption, pain, fever, systemic inflammatory response syndrome (SIRS) and sepsis.**
- Measuring IRP has needed a **separate sensor** in the collecting system; the **LithoVue Elite SDFU** integrates a pressure sensor for **continuous real-time monitoring**, which should allow **safe thresholds and guidelines** to be defined. Temperature and fragment-size sensing are expected next.

Concept — Every ureteroscope choice trades **calibre against image and durability against cost**: the fibre bundle gives the smallest tip but degrades, the chip gives the best image but a larger tip and no durability gain, and single use removes repair cost at a price that only low-volume centres recoup.

CONSENT AND ACCESS: URETEROSCOPY

- **Consent covers:** failure to access the ureter or kidney; incomplete fragmentation or clearance; further procedures; **perforation, avulsion and sepsis**; and long term, **stricture, further surgery, loss of renal function or nephrectomy**.
- **The first step is to identify the ureteral orifice and place a safety wire**, which may also serve as the working wire.
- **Segura's dictum:** the only difference between an endourological procedure and a motorcycle accident is **the presence of a safety wire**. Its routine value is debated, but in the rare case that needs it, it is decisive.

Getting to the Stone

- The "difficult ureter" is **narrow or tortuous** and will not admit instruments easily. The choice is between endourological manoeuvres now and a **stent for passive dilation with a staged return** — the latter costs extra procedures, anaesthetics and convalescence but is **among the safest options**.
- **Second-look URS after pre-stenting is best done after 14 days and within 5 weeks**; earlier or later procedures had lower SFRs.

Ureteral Access Sheaths

- **UASs suit proximal ureteral and renal stones best.** For renal stones they may improve SFR and **lower infection risk, irrigation pressure, scope wear and operative time**.
- **For ureteral stones the sheath helps when there is a large burden requiring repeated scope passes**, when **dusting relies on irrigation outflow for vision**, and when high laser energy creates **heat that must be carried away**.
- **Placement uses two wires:** the safety wire holds access while the **sheath passes over a second, stiff working wire under fluoroscopy**. Sheaths have a **hydrophilic coating and a tapered obturator**; the obturator may be passed **alone first** to dilate.
- **A safety wire outside the sheath can cause resistance:** either use the safety wire as the working wire, or use sheaths designed to keep the wire outside the lumen, where it still serves as a safety wire.
- **About half of endourologists use a UAS for ureteral stones, most commonly 12/14 Fr.**
- **Sheath pressure can cause ureteral ischaemia and stricture;** in animals **14/16 Fr sheaths reduced ureteral blood flow faster, more deeply and for longer** than smaller ones. Clinically, stricture after sheath

use is rare; **earlier high stricture rates reflected larger scopes and repeated passage with fragments**, the very trauma a sheath prevents.

- **Sheaths are hard to stabilise low in the ureter**, so most reserve them for **stones above the iliac vessels**, with the tip **just below the stone**; excess external length is unstable and **shortens the scope's working length**.

Guidewire Types

- Wires range from **0.018 to 0.038 inch in diameter and 145 to 260 cm in length**, in varied materials, stiffness and coatings.
- **Tactile feedback and judicious fluoroscopy guide passage**, following **ALARA (as low as reasonably achievable)** but never at the expense of safety or the surgical goal.
- **If the wire's path is uncertain, pass an open-ended or dual-lumen catheter and perform a retrograde pyelogram (RPG)**; whether an RPG should be routine before every URS is debated.
- **Nitinol** (nickel–titanium) is **pseudoelastic with shape memory and resists kinking**; polytetrafluoroethylene (PTFE) wires with a nitinol core are very kink-resistant. The aim is a wire **stiff enough to carry instruments yet flexible enough to pass an obstruction**.

GUIDEWIRE CLASSES — COMPARISON

	Hydrophilic	PTFE-coated	Hybrid
Surface; core	Lubricious when wet; nitinol	Hydrophobic; stainless steel or nitinol	Floppy hydrophilic nitinol tip, stiff PTFE stainless-steel shaft
Best for	Bypassing obstructing or impacted stones with low perforation risk	Safety or working wire; resists displacement ; eases passage of instruments	Difficult anatomy plus instrument passage; some have scope-protecting segments
Weakness	Slippery — easily displaced from the ureter	Damages the flexible-scope channel on backloading	—

Other enhancements include **markings, removable cores to vary tip flexibility, and steerability**. The trap is the pairing: the hydrophilic wire is the best for getting past a stone but the worst at staying in place.

Safety Wires

- **A safety wire delineates anatomy, prevents loss of access and allows a stent after perforation**.
- **"Wireless" URS**, using the flexible scope itself as the safety wire, has produced **low complication rates without perforation or avulsion in uncomplicated cases**.
- **Because complexity cannot be predicted in advance, most experts still advocate a safety wire**, tailored to the case and the surgeon.

Obstructed Ureters

- **About one ureter in ten cannot be accessed safely** despite miniaturisation. Single-stage techniques spare anaesthetics, but **safety governs: a double-J stent and a staged procedure are preferred to risking injury**.

Wire will not bypass the stone. Open-ended catheter over the wire to **1–2 mm short of the obstruction** → **RPG** to outline the obstruction and any contrast passing it → using contrast **hydrodissection**, pass a **hydrophilic floppy-tip wire** at once, the catheter's stiffness directing the force → if successful, **advance the catheter past the obstruction** → **exchange for a stiff wire**, which also straightens a tortuous ureter.

- **If the catheter abuts the stone, the wire cannot find the natural channel and may exit at 90° — perforation**. A torque vice and angled wires or catheters help in a tortuous ureter.
- **Alternative:** leave the wire at the obstruction as a safety wire, **displace the stone or perform direct lithotripsy**, then place the safety wire **under direct vision** once the proximal ureter is seen.

Ureteral Dilating Balloons

- **Balloons (4–10 cm) pass over a wire under fluoroscopy.** There is no set inflation time; the working rule is **pulsed fluoroscopy until the waist resolves → hold 1–2 minutes → deflate and remove**, the wire remaining for the stent.
- **Miniaturisation has made active orifice dilation uncommon;** even an early 7.5 Fr flexible scope needed none in about half of cases.
- **When the wire is in but the scope will not enter the orifice, persistence risks a mucosal flap, false lumen and perforation.**

Scope will not enter the orifice.

- **Railroading:** a second wire is placed through the scope into the ureter, and the scope **dilates its way in between the two wires; best with a rigid or semirigid scope.**
- **Dual-lumen catheter:** dilates the ureter and places a **second wire or performs an RPG** through its second channel; the scope then passes between the wires.
- **Rotate the flexible scope,** because its **working channel is offset** from the centre of the tip; rotation minimises the leading profile.
- **Use the UAS obturator as the dilator** (Table 97.1), saving a disposable dilator and avoiding a dual-lumen catheter.

TABLE 97.1 INSERTING A SECOND GUIDEWIRE USING A URETERAL ACCESS SHEATH (UAS)

Obturator alone over the guidewire to test and dilate → **obturator + sheath into the distal ureter** → remove the obturator → **second guidewire through the sheath** (position confirmed when external lengths match, and by fluoroscopy) → **remove the sheath**, leaving two wires → **reload obturator and sheath over one wire** (a stiffer wire helps), then remove that wire and the obturator → **the external wire is the safety wire.**

The sequence creates a safety wire outside the sheath using only the sheath's own obturator. The trap is the end state: the wire inside the sheath is removed, and only the external wire remains.

TABLE 97.2 METHODS TO DILATE A URETER

Method	Best for	Note
Indwelling stent (passive)	Whole ureter narrow	Gentlest ; staged return. For avascular strictures , balloon may be preferable
Serial dilators	Long narrow segments	Progressively larger sizes
Coaxial dilators, 8–10 Fr	General dilation	Also allow RPG and safety wire
Balloon	Focal (short) stenosis	Wire stays in place
UAS obturator, then sheath	Planned sheath use	Dilates and gauges tightness
Semirigid ureteroscope	Distal ureter	Dilates while inspecting

The method is matched to the **length of narrowing**: diffuse → stent or serial dilators; focal → balloon. The trap is that the gentlest method commits the patient to a second procedure.

Pre-stenting. A pre-placed stent dilates the ureter passively, makes most of these manoeuvres unnecessary, and **reduces ureteral injury during sheath insertion.**

PRE-STENTING BEFORE URS — EVIDENCE SERIES

Study type	Finding
Retrospective matched pair	Higher SFR for ureteral stones, confined to stones ≥5 mm

Study type	Finding
Meta-analysis	Better SFR for renal stones and flexible URS; no effect for ureteral stones or semirigid URS
Large global URS database	No SFR advantage for ureteral stones

Verdict: pre-stenting is not necessary for URS of ureteral stones; the positive retrospective result is outweighed by the meta-analysis and the database.

α-Blockers Before Ureteroscopy

- **The rationale is the abundance of α1 receptors in the distal ureter**, which govern peristalsis and tone.
- A meta-analysis of randomised trials found **less need for intraoperative dilation, better access to the stone, higher stone-free status at 4 weeks and shorter operative time**. α-Blockers may also **ease postoperative and stent pain and lower the force needed to insert a 12/14 Fr sheath**.
- **Without adequately powered trials, use is judicious** — distal stones, or when large scopes or a sheath are expected. **Most trials gave 1 week preoperatively**.

Concept — Every access manoeuvre is a way of **widening the ureter without tearing it**: the hierarchy runs from time (stent, α-blocker) through graded mechanical dilation to direct force, and safety always permits retreating to a stent and a staged procedure.

LITHOTRIPSY TECHNIQUES

Basketing Techniques

- **A small stone may simply be grasped and removed**; retrieval devices vary in shape, size, material and engagement ability.

Engaging from behind. Pass the basket **beyond the stone** → **open** → draw back toward the scope **under vision** → **agitate gently** → **close slowly**. For a stone that drifts forward, open the basket **just larger than the stone** and **push it slightly forward as it closes**.

- **Keep the stone 5–10 mm from the scope tip** during withdrawal so the ureter is always seen and not caught; **blind basketing is unsafe**.
- **Align the stone's long axis with the ureter** by partly opening and agitating the basket, using the ureteral walls or irrigation.

Entrapped stone. Open the basket and **push the stone into a wider segment** → if immovable, **disassemble the basket**, leaving its shaft as a wire → **reinsert the scope alongside** → **laser the stone** (or a nitinol basket) → if the basket cannot be disassembled, **cut it distal to the handle**.

- **Dual-channel scopes** allow a small fibre through the irrigation channel without disassembly; **very small baskets (1.5 Fr Sacred Heart halo)** allow laser and basket together in a single channel.
- **Assess before engaging; when in doubt, fragment before extracting**, to prevent perforation and avulsion.

Electrohydraulic Lithotripsy

- **Electrohydraulic lithotripsy (EHL)**, described in the 1950s, **discharges a spark through fluid**, creating an expanding vapour bubble and a pressure wave that radiates **in all directions** — hence its danger.
- The **3–5 Fr probe** limits irrigation, must sit **on or very near the stone** and **away from the scope tip**; hard stones (**cystine, calcium oxalate monohydrate**) resist it, and **ureteral trauma risk is high**, so an **RPG after treatment** checks for extravasation.
- **Holmium laser proved more effective for larger stones and faster**, so EHL has largely been abandoned.

Pneumatic Lithotripsy

- **Compressed air drives a metal probe against the stone like a jackhammer.** The probes are semirigid, so they work **only through rigid or semirigid scopes — never flexible ones**; the usual **base rate is 12 Hz**, and **flexing the probe off axis wastes energy**.
- **Success is greatest below the pelvic brim and least above it**, and falls with **larger stones and existing hydroureteronephrosis**.
- The energy is not directly traumatic, but **pressing the probe on a partly fragmented stone can perforate the ureter**. The chief limitations are **accessibility** and **proximal stone migration**.

Antiretropulsion devices. Placed **between stone and kidney** to stop fragments migrating, they may double as a safety wire but add cost and time. **Migration matters most with rigid or semirigid URS**, where it may end the procedure or force conversion to antegrade percutaneous access or SWL. They improve SFR, but **demand has fallen with flexible URS**.

FIG. 97.3 ANTIRETROPULSION DEVICES

Device	Mechanism
Stone Cone (Boston Scientific)	Conical wire forming a cone behind the stone
Accordion (Accordion Medical)	Film passed flat; pulling its wire folds it into a barrier
NITrap (Cook Medical)	Nitinol catch behind the stone

Holmium:YAG Laser Settings

- **Early low-power, short-pulse (SP) holmium:yttrium-aluminium-garnet (Ho:YAG) lasers fragmented stones**, the short pulse breaking pieces off, which were split further until they could pass or be basketed.
- **Higher-power Ho:YAG introduced dusting — low pulse energy at high frequency.** Fragments can then be "popcorned": they whirl in a cyclone-like motion and are ground to dust as they meet the laser (Fig. 97.4).
- **Lower pulse energy produces smaller pieces and less retropulsion**, so the stone is easier to target continuously; **long-pulse (LP) and pulse-modulated Ho:YAG** reduce retropulsion further. **The fibre tip serves as a size reference** for fragments.
- **High dusting settings (0.3 J × 80 Hz or 0.2 J × 120 Hz, both 24 W) heat the irrigant**; for large stones needing prolonged lasing, **a sheath for outflow and room-temperature or colder irrigant** are needed, and **in the ureter power should be capped at 10 W**.

Thulium Fibre Laser Settings

- **The thulium fibre laser (TFL) can fire above 2000 Hz**, but a randomised trial showed **lower settings may be optimal**: TFL was **faster than Ho:YAG**, **equal for ureteral SFR** and **better for renal zero-fragment rate**, challenging the idea that low energy with high frequency is always best.
- **Why lower settings work**: power (W) is heat, so low settings **widen the safety margin**; lower frequency means **more energy strikes the stone rather than water** and **less misfires into tissue**. **Start low; increase only as needed**.
- **TFL's lower peak power and long pulse (about 50–200 μs) ablate stone into fine particles** rather than breaking off fragments, so technique differs from SP Ho:YAG.
- **In an impacted ureteral stone, target the centre first**: as space opens the sides fall into the lumen and can be lasd **away from the wall**; the scope tip dislodges fragments stuck to urothelium. Increase energy in **small steps**, keeping **total power ≤10 W**; fragments are usually small enough that **no basket is needed**.
- **If visibility is poor, lower the frequency and keep power constant by raising energy**.
- **In the kidney, ureteral settings are safe, or dusting settings can be used**; **paint the stone surface and do not drill** — with TFL drilling melts the stone and causes flashing, although it can fragment with SP Ho:YAG.

Above 40 Hz many pulses miss the stone and strike urothelium. SP duration is used throughout, since TFL retropulsion is already minimal and longer pulses add nothing.

LASER SETTINGS — HO:YAG VERSUS TFL

	Ho:YAG	TFL
Stone effect	SP breaks off larger fragments ; drilling can fragment	Fine particles ; drilling counterproductive
Retropulsion	SP high; LP and pulse-modulated lower	Minimal, even at SP
Fragmentation settings	0.6–1.0 J × 6–10 Hz (3.6–10 W)	Start 0.6 J × 6 Hz SP (3.6 W) ; rarely >1.0 J
Dusting settings	0.2–0.3 J × 40–80 Hz (8–24 W)	0.2 J × 40 Hz (8 W); >40 Hz often unproductive
Heat	Identical rule for both: heat tracks power, not laser type	

Laser Settings for Ureteroscopy

- Settings depend on **location and desired outcome** — fragment and basket, or dust and let pass — and **both yield high-quality results**.
- A second randomised trial compared a 60 W TFL with a 120 W pulse-modulated Ho:YAG at **higher starting settings** (fragmentation 0.8 J × 8 Hz, 6.4 W; dusting 0.3 J × 80 Hz, 24 W) and found **SFR, operative time, complications and zero-fragment rate similar**, though follow-up used **KUB and US rather than the more sensitive NCCT**.
- **Criticism:** the manufacturer's Ho:YAG dusting setting **may not suit TFL**, since at high frequency fewer pulses hit the stone and more energy is absorbed by fluid.

RANDOMISED TFL VERSUS HO:YAG TRIALS — COMPARISON

Trial	Settings	Verdict
First randomised trial	Low settings	TFL faster; equal ureteral SFR; better renal zero-fragment rate
Second randomised trial	Higher settings	Equivalent; less sensitive imaging

The divergence is explained by the settings: high-frequency dusting may handicap TFL, so the setting chosen, not only the laser, decides the comparison.

Temperature Safety When Using Lasers

- **Heat is a by-product of every laser**; attention to settings and **thermal relaxation time** prevents **delayed thermal injury and stricture**.
- **The ureter and a calyx hold small fluid volumes that heat quickly; above 43 °C, tissue is damaged.** Without irrigation, every setting tested on the bench exceeded 43 °C.
- **Laser type does not determine heat — power, technique and adjuncts do.** Fluid temperature tracks power for both TFL and Ho:YAG, so the rule is **the lowest effective power**, with a guide of **≤10 W in the ureter and ≤25 W in the kidney**.
- **An access sheath lowers both IRP and temperature** by providing outflow, with both lasers; in an animal model, a **14 Fr sheath kept temperature below 44 °C** where no sheath did not.
- **Low-retropulsion lasers keep the stone still, allowing continuous lasing and so more heat**; how the laser is used matters more than where.

- **Irrigant should be room temperature or colder and run continuously.** The old habit of **stopping irrigation to limit retropulsion** overheats the ureter and **risks thermal injury and stricture.**

Concept — Power is heat and heat is stricture: every laser recommendation — low settings, 10 W in the ureter, 25 W in the kidney, a sheath for outflow, cool continuous irrigation — follows from keeping energy delivered to a small fluid volume below 43 °C.

DRAINAGE AND ANTIBIOTICS AFTER URETEROSCOPY

- Two decisions close every case — **stent or not, and postoperative antibiotics or not** — and the evidence below applies to **uncomplicated URS** without infection, impaction or perforation.

Antibiotics

- **Postoperative antibiotics are indicated only if the patient had urosepsis during that stone episode.** A **single perioperative intravenous dose** gives the same infection rate as a postoperative course, so the default is **one dose and nothing afterwards** unless something arises during the case.
- **Higher infection risk:** higher Charlson comorbidity index, older age, female sex, **prolonged pre-stent dwell time**, neurogenic bladder and high body mass index (BMI).

TIMING OF DEFINITIVE URS AFTER PRE-STENTING — TWO WINDOWS

Endpoint	Window
Stone-free rate	After 14 days, within 5 weeks
Postoperative sepsis	2–4 weeks

Different endpoints give overlapping windows; **2–4 weeks satisfies both.**

Ureteral Stents

- **Stents drain and dilate the ureter for passage of fragments and dust, but are a major source of morbidity — symptoms in about four in five patients.**
- A randomised trial (Denstedt et al., 2001) showed URS can be performed safely without a stent, and a Cochrane review found pain, return to hospital and secondary interventions similar with or without one.
- **Optimal stent duration is unknown;** removal at 3 rather than 7 days caused **more unplanned returns and flank pain**, with no difference in hydronephrosis.
- **Removal by string equals cystoscopy for infection and outcome, is preferred by patients and is cost-effective.**
- **Degradable stents** would need no removal and may be better tolerated if biocompatible and **not inhibiting peristalsis**, which is thought to contribute to stent symptoms.
- **Access-sheath use does not affect postoperative hydronephrosis** at 8 weeks.
- **Practice lags evidence:** more than half of urologists stent routinely, and about three-quarters in the United States.

POSTOPERATIVE STENT — REQUIRED VERSUS DEBATABLE

Required	Decide at the end of the case
Intraoperative perforation; stricture requiring dilation; tight ureter; signs of infection; planned second-stage URS	After sheath use — once routinely stented, now challenged: inspect the ureter after sheath removal. Pre-stented + sheath + normal ureter → no stent.

Concept — After uncomplicated URS the evidence runs one way — **one dose of antibiotic and no stent — and a stent is reserved for a named indication, because otherwise it adds morbidity without adding safety.**

SHOCK WAVE LITHOTRIPSY FOR URETERAL CALCULI

- **SWL remains a worldwide mainstay and the only truly effective noninvasive treatment**, though **burst wave lithotripsy (BWL)** may change that.
- Modern machines treat most ureteral stones but **may need more than one session**; SWL can be done under **intravenous sedation or local anaesthesia**, and patients value **avoiding a stent and general anaesthesia (GA)**.
- **Quality-of-life (QoL) comparisons may favour SWL**, but most used **generic instruments** that miss the disease-specific impact.

Indications for Shock Wave Lithotripsy for Ureteral Calculi

- **SWL is an option for a stone anywhere in the ureter**; because outcome depends on stone and patient factors, **informed consent and shared decision-making**, including patient preference, are central.
- In acute colic with an obstructing stone, **SWL within 48 hours outperforms delayed SWL**; emergency SWL gives **similar SFR in the upper, mid and distal ureter**, low complication rates, and rarely needs urgent drainage.
- **Distal ureteral stones are treated safely and effectively**, despite some centres' reluctance, with a **second session adding clearance**.

GUIDELINE POSITIONS ON SWL FOR URETERAL STONES — COMPARISON

Body	SWL in all ureteral locations	Single-treatment SFR
European Association of Urology (EAU), 2023	Option in all locations — all agree	Higher with URS, particularly for larger stones
American Urological Association (AUA) / Endourological Society, 2016		
Canadian Urological Association (CUA), 2015		

Contraindications to Shock Wave Lithotripsy for Ureteral Calculi

CONTRAINDICATIONS TO SWL — ABSOLUTE VERSUS RELATIVE

Absolute	Relative
Pregnancy; uncontrolled bleeding disorder; anticoagulation that cannot be paused for treatment and 48 hours after; untreated anatomical obstruction distal to the stone; uncontrolled or untreated UTI; untreated or severe hypertension	Skeletal abnormality or severe obesity preventing targeting or placement in the focal zone; adjacent arterial aneurysm

Patient Selection

- **Stone size is the most important predictor of SWL outcome**, and the **largest linear dimension predicts as well as area or volume**. SFR is slightly lower than URS overall, and **the gap widens in favour of URS at ≥ 10 mm**.
- **Location matters little once size is accounted for**.
- **Pure cystine and dense, smooth, homogeneous stones (brushite, some calcium oxalate monohydrate) fragment poorly**, although composition is often unknown before treatment.
- **Chronic impaction worsens outcome and may be the strongest predictor of all**. CT suggests it by **soft-tissue density of the ureteral wall around the stone (the "ring" sign) with severe hydronephrosis**, or by a stone **unchanged in position on earlier imaging**.
- **Hounsfield units (HU) above 1000 and SSD above 10 cm predict failure for renal stones**; for ureteral stones the evidence is mixed, and they may add nothing once size and impaction are considered.

System Factors

- Where SWL is **unavailable, distant or offered only on certain days, local URS may be preferable.**

Patient Preference

- **An informed patient may choose SWL despite a predicted inferior result** — a stone over 10 mm, high HU or some impaction — accepting a possible second session or salvage URS.
- **When rapid clearance in one treatment is critical** (imminent travel, return to work or caregiving), **URS may be preferred.**

Patient Preparation

- **Routine pre-stenting before SWL is not needed:** it improves neither SFR nor retreatment, though SWL can proceed with a stent already in place. **A stent may reduce steinstrasse after large stones but does not change its presentation or management, and stents worsen lower urinary tract symptoms (LUTS).**
- **Correct a bleeding diathesis first or choose URS.** If thrombotic risk allows, **hold acetylsalicylic acid (ASA), non-steroidal anti-inflammatory drugs (NSAIDs) and anticoagulants to achieve normal coagulation on the day and for 48 hours after,** especially for upper ureteral stones where **kidney, spleen or liver lie in the blast path.** Continuing **low-dose aspirin (81 mg) or NSAIDs** has been reported for mid and distal stones.
- **Hypertension increases bleeding and perinephric haematoma** when the kidney is in the blast path; untreated or severe hypertension is an absolute contraindication, and **even controlled hypertension raises risk.**
- **Pacemakers and defibrillators may need deactivation or reprogramming,** particularly for upper ureteral stones near the cardiac conducting system.
- **Urinalysis and culture before SWL; treat any bacteriuria first.** Routine prophylaxis is unnecessary, but antibiotics are given with **indwelling stents, nephrostomies or catheters, struvite stones, recent or recurrent UTI, and colonised or reconstructed tracts** (ileal conduit, augmentation, neobladder).

Technique of Shock Wave Lithotripsy to Maximise Outcomes

- **Good analgesia or anaesthesia keeps the patient still, and GA or intravenous sedation outperforms minimal anaesthesia,** because **stone movement of 10–15 mm significantly impairs fragmentation.** Target during **expiration, two-thirds of the respiratory cycle,** and use **pressure belts.**
- **Position by bone:** upper ureter **supine;** mid-ureteral stones over the sacrum or pelvis **prone,** so bone does not absorb the energy; stones below the sacroiliac joint **supine.**
- **Coupling is critical, especially with "dry" shock heads,** because air dissipates shock waves: the gel, water or oil interface must be **free of bubbles,** as even tiny air pockets sharply reduce energy transfer.
- **Shock rate is one of the most important parameters: 60–90 shocks/min (1–1.5 Hz) outperforms 120 (2 Hz)** in randomised trials and meta-analyses, and the slower rate also **protects the kidney and its vasculature.**
- **Treat until complete fragmentation or the manufacturer's maximum shock count,** with **continuous or frequent imaging.** US and fluoroscopic targeting are **equally effective; longer fluoroscopy time predicts better outcome,** reflecting better targeting.
- **Ramp energy gradually to the maximum:** this improves fragmentation and **reduces renal injury** by inducing protective vasoconstriction and lowering injury markers (**microalbumin, β 2-microglobulin**).
- **Experienced urologists and technologists, and ongoing quality assurance, give better results.**

- **After a failed session**, choose between repeat SWL and **salvage URS** by the degree of fragmentation and patient priorities; **benefit beyond two sessions for the same stone is small**, and the interval can be as short as **1–2 days** for mid and distal stones.

Adjuncts to Improve Shock Wave Lithotripsy Outcomes

ADJUNCTS AFTER SWL — COMPARISON

Adjunct	Effect	Qualification
α-Blockers	Higher SFR, faster clearance by relaxing ureteral smooth muscle	Particularly larger stones
Paracetamol + NSAIDs	Opioid-sparing analgesia	—
Potassium citrate or alkali	Higher SFR for calcium oxalate, uric acid and mixed stones	Mainly renal data
Forced diuresis	No benefit	—

Comparative Data of Shock Wave Lithotripsy Versus Ureteroscopy

SWL VERSUS URS FOR URETERAL CALCULI — COMPARISON

Domain	SWL	URS
SFR	Lower after one treatment; gap widens ≥ 10 mm or impacted	Higher at 2 weeks; equal at 3 months
Further treatment	More retreatment ; fewer auxiliary procedures and stents	Less retreatment; more stents → more pain and analgesia
Complications	Lower, including UTI	Ureteral perforation
Anaesthesia; recovery	Sedation or local; shorter stay and recovery	Usually GA; longer
Cost	Lower, even with two sessions	Higher

- **Salvage URS after failed SWL for a ureteral stone is no harder and no less successful** — unlike salvage after failed SWL for renal stones.
- **QoL evidence is mixed**: URS fares better when stentless or emergency, SWL when URS is followed by a stent; studies are small, unblinded and use generic instruments.
- **SWL remains safe and effective for virtually all ureteral stones**, allowing most patients to be treated without GA or stent and with less morbidity, at the cost of slightly lower single-treatment SFR for stones ≥ 1 cm or impacted.

Concept — SWL and URS converge on the same end point by different routes: **URS buys speed and single-treatment certainty with invasiveness and a stent**; **SWL buys low morbidity at the cost of retreatment, and size, impaction and the patient's priorities decide which price to pay.**

SPECIAL SITUATIONS

Pregnancy

- **Conservative management is first-line** if safe — **no infection, oliguria, anuria, solitary kidney or fetal distress** — and **SWL is contraindicated**.
- **If conservative treatment fails, URS is safe**, with high success and near-universal normal continuation of pregnancy; sedation, GA, spinal or epidural anaesthesia add no risk, and **limited fluoroscopy with fetal shielding** is safe.
- **Stents are safe but encrust faster in pregnancy**, so they are **changed every 4 weeks**, which shapes the timing of definitive treatment. **Retrograde stent is attempted before nephrostomy.**

Urinary Diversion and Transplanted Kidneys

- After diversion, reimplantation or transplantation, retrograde access is the hardest part; flexible cystoscopy through an ileal conduit or **Indiana, Kock or Studer pouch** is often very difficult or impossible.
- **SWL can be tried**, allowing for altered positioning — **more often prone** — where the ureter crosses bone.
- **The most definitive route is antegrade and percutaneous**; guidelines support percutaneous treatment of large proximal ureteral stones even without diversion. **Antegrade flexible URS through a percutaneous sheath** benefits from a **dilated proximal ureter** and lets **dust wash down into the bladder or diversion**.
- In transplanted kidneys both antegrade and retrograde approaches are safe and successful.

Congenital Renal Anomalies

- In ectopia, crossed-fused ectopia and complete or partial duplication, retrograde URS is usually feasible with good results; the challenge is an **ectopically placed orifice**, as in some duplex systems.
- **Escalation**: SWL → PCNL if SWL fails or retrograde access is impossible → **laparoscopic ureterolithotomy or pyelolithotomy** as salvage.

Concept — When the retrograde route is compromised — by pregnancy's limits on radiation and SWL, or by reconstructed anatomy — **the choice shifts to whatever route is safest: stent and URS in pregnancy, antegrade access after diversion, laparoscopy as the final salvage.**

COMPLICATIONS

- Complications are becoming rarer with smaller scopes, better vision and efficient lasers; most are **minor** (perforation) and easily managed (stent), and the major one, **avulsion**, is very rare and preventable.

Perforation

- Incidence 0–4%, and falling. **Causes**: the scope; forcing an oversized sheath; unseen distal slippage of the dilating wire during sheath placement (hence fluoroscopy); **laser on the wall**; lithotrite or electrocautery, which cause full-thickness injury; stone manipulation; **wires without a floppy tip**; baskets gripping the wall; and **poor visibility** from bleeding or debris.
- If a procedure is difficult and not progressing: **stop, stent, stage**.

Management. **Stop** → **stent across the injury** (no further lithotripsy, since fragments in the wall provoke a **granulomatous reaction and stricture**) → **saline irrigant** to prevent electrolyte disturbance from extravasation → **drain a urinoma and give antibiotics** → **stent for about 4 weeks** → **mandatory imaging after removal**.

Submucosal Stone

- Submucosal and "lost" stones are more common with **chronic obstruction**; a submucosal stone is **lasered free and a stent placed**, because of the risk of **later stricture**.
- A stone fully extruded through a perforation is left where it lies — retrieval worsens injury and extravasation. **Stop, stent, image afterwards, and tell the patient**, since it may later be mistaken for a new ureteral stone.

Avulsion

- The most disastrous complication, occurring in about 0.06–0.5% of cases, it results from **forceful manipulation of a large or impacted stone**. The **proximal third** is most at risk, having the **least muscular support**, and the injury is recognised when **ureter emerges with the stone and basket**.
- **Prevention**: never basket blind; keep space between stone and scope; **fragment stones too large to remove intact**; release or reposition an oversized engaged stone, or **disassemble the basket and fragment in situ**; a **safety wire allows a stent and abandonment at any time**.

- **Management: nephrostomy and delayed repair, not immediate primary repair;** drain any urinoma; reconstruction depends on **level and remaining viable ureter**. **Nephrectomy is not recommended**, because stone disease recurs and stone formers are prone to **hypertension and diabetes**.

Stricture

- **Incidence 3–6%, falling with better technology and technique. Risk factors: impacted or long-standing stones, perforation, previous ureteral surgery and pelvic radiation** (poor blood supply and healing), though some strictures follow uncomplicated URS.
- **Avoid aggressive passage through a narrow segment, and image every patient after URS**, since some strictures cause **silent obstruction**.
- **Management depends chiefly on length and location**, plus time since injury, mechanism and patient factors.

Post-Ureteroscopy Sepsis

- **Sepsis follows about 5% of URS in meta-analysis and is serious**, bringing longer stay, **intensive care unit (ICU)** admission, readmission, cost and **mortality**.
- **Risk factors: positive preoperative culture, preoperative stent, older age, diabetes, ischaemic heart disease (IHD) and longer surgery**.
- **Prevention: culture and treat before surgery; keep IRP as low as possible** in at-risk patients. **Treatment: broad-spectrum antibiotics and support** until urine, blood or stone cultures identify the organism.

Concept — Nearly every serious URS complication traces to **force applied without vision** — *blind basketing, forced sheaths, persistence in a tight ureter* — so the cure is uniform: **see, fragment, and when in doubt stop, stent and stage**.

FUTURE OF URETEROSCOPY IN URETERAL STONES

- **Smaller scopes, digital chips and temperature and pressure sensors** are already here, and further miniaturisation will ease access.
- **Surgeon ergonomics and repetitive strain injury** are driving **robotic URS**, which adds degrees of freedom as it did in laparoscopy.
- **Artificial intelligence (AI)** may adjust laser settings from **fragmentation feedback**, control **irrigation, temperature and IRP**, compensate for **respiration**, and navigate by **electromagnetic tracking** integrating fluoroscopy, preoperative CT and sensor data.
- **BWL (Break Wave, SonoMotion)** uses a **handheld ultrasound device** delivering low-amplitude ultrasonic bursts that make the stone **resonate, vibrate and shatter**. Preliminary data show it **fragments all stone types in the ureter**; it is expected to be an **adjunct to SWL and URS**, suited to certain locations and SSDs.

ROBOTIC URETEROSCOPY PLATFORMS — COMPARISON

Type	Benefit
Robot holding a standard commercial ureteroscope	Ergonomics; surgeon away from fluoroscopy, so less radiation
Robot with a dedicated integrated ureteroscope	Tighter integration, including electromagnetic location sensing

SWL VERSUS BWL — COMPARISON

	SWL	BWL
Energy	High-amplitude shock waves	Low-amplitude ultrasonic bursts; resonance
Anaesthesia	Sedation or local	Little or none
Setting	Lithotripsy suite	Clinic or emergency department

CONCLUSION

- **No single formula fits every ureteral stone:** SWL, URS and PCNL are all options, chosen individually through **informed decision-making**, and new noninvasive treatments are close.